N20 Chemical reaction rate

Bohaz

21.10.2026

1

This note summarises the key results in Prof. D. E. Manolopoulos FRS lecture series on 'Chemical Reaction Dynamics' for the MSc in Theoretical and Computational Chemistry course in Oxford. We will also provide a few alternative and simpler proves to certain parts.

We will consider a simple 1D reaction. There is a potential barrier $V(\hat{x})$ which vanishes as $x \to \pm\infty$, a particle with reaction coordinate $x < s$ is on the reactant side and $x > s$ is on the product side. The Hamiltonian of the problem is

$$\hat{H} = \hat{H}_0 + V(\hat{x}) = \frac{\hat{p}^2}{2m} + V(\hat{x}) \tag{1.1}$$

the scattering eigenstates with energy E satisfy the eigenvalue problem above

$$\hat{H}|\psi_p\rangle = E|\psi_p\rangle \tag{1.2}$$

the asymptotic behaviour of the scattering state in the reactant side is

$$|\psi_p(x \to -\infty)\rangle \sim |p\rangle + |-p\rangle R(E) \tag{1.3}$$

where $R(E)$ is the reflection amplitude, and asymptotic behaviour in the product side is

$$|\psi_p(x \to \infty) \sim |p\rangle T(E) \tag{1.4}$$

where $T(E)$ is the transition probability. Probability of all event must add up to 1, so we have

$$|R(E)|^2 + |T(E)|^2 = 1 \tag{1.5}$$

The Møller operator $\hat{\Omega}_+$ is defined as

$$\hat{\Omega}_+ = \lim_{t\to\infty} e^{-i\hat{H}t/\hbar} e^{+i\hat{H}_0 t/\hbar} \tag{1.6}$$

which connects the free state to a scattering state

$$|\psi_p\rangle = \hat{\Omega}_+ |p\rangle \tag{1.7}$$

which evolves a free state to the distant past, then match it onto a scattering state and evolve forward in time to present. Now introduce the (product) side operator

$$\hat{h} = h(\hat{x} - s) \tag{1.8}$$

where h is the unit step function, its derivative is a Dirac delta with a spike at where the step function jumps from 0 to 1. The reactant side operator is $1 - \hat{s}$. Define the flux operator as the Heisenberg derivative of side

$$\hat{F}(s) = \frac{i}{\hbar}[\hat{H}, \hat{h}] \tag{1.9}$$

The product side operator is a projector into the subspace of product

$$h(\hat{x} - s) = \int_{-\infty}^{\infty} dx\, |x\rangle h(x - s)\langle x| = \int_{s}^{\infty} dx\, |x\rangle\langle x| \tag{1.10}$$

and similarly the reactant side operator is a projector into the subspace of reactants. One can show that

$$\hat{F} = \frac{1}{2m}\left(\hat{p}\delta(\hat{x} - s) + \delta(\hat{x} - s)\hat{p}\right) \tag{1.11}$$

then note

$$\delta(\hat{x} - s) = \int_{-\infty}^{\infty} dx\, |x\rangle\delta(x - s)\langle x| = |s\rangle\langle s| \tag{1.12}$$

therefore

$$\hat{F} = \frac{1}{2m}\left(\hat{p}|s\rangle\langle s| + |s\rangle\langle s|\hat{p}\right) \tag{1.13}$$

Then

$$\langle\psi_p|\hat{F}|\psi_p\rangle = \langle p|\hat{F}|p\rangle|T(E)|^2 = \frac{p}{m}|\langle s|p\rangle|^2|T(E)|^2 = \frac{1}{2\pi\hbar}\frac{p}{m}|T(E)|^2 \tag{1.14}$$

2

The thermal rate coefficient is

$$k(T) = \frac{1}{2\pi\hbar Q_r(T)}\int_0^{\infty} dE\, e^{-\beta H} N(E) \tag{2.1}$$

where $\beta = 1/k_B T$ abd $Q_r(T)$ is the partition function per unit length

$$Q_r(T) = \frac{1}{\Lambda(T)} = \left(\frac{2\pi m k_B T}{h^2}\right)^{1/2} \tag{2.2}$$

$N(E)$ is the cumulative reaction probability, in our simple 1D case this is

$$N(E) = |T(E)|^2 \tag{2.3}$$

Then through a rather clever series of algebraic manipulation in Eq.(85)-(88) in the chemical reaction dynamics notes, we have

$$k(T)Q_r(T) = \lim_{t\to\infty} \mathrm{tr}\left[e^{-\beta\hat{H}}\hat{F}e^{-\beta\hat{H}}\hat{\Omega}_+ h(\hat{p})\Omega_+^{\dagger}\right] \tag{2.4}$$

Consider results in Appendix A, we have

$$e^{-i\hat{H}_0 t/\hbar}h(s - \hat{x})e^{+i\hat{H}_0 t/\hbar} = h(s - e^{-i\hat{H}_0 t/\hbar}\hat{x}e^{+i\hat{H}_0 t/\hbar}) = h\left(s - \hat{x} + \frac{\hat{p}t}{m}\right) \tag{2.5}$$

then

$$\lim_{t\to\infty} e^{-i\hat{H}_0 t/\hbar}h(s - \hat{x})e^{+i\hat{H}_0 t/\hbar} = h\left(\hat{p} + \frac{m(s - \hat{x})}{t}\right) = h(\hat{p}) \tag{2.6}$$

substitute this back into Eq.(2.18) gives

$$k(T)Q_r(T) = \lim_{t\to\infty} \mathrm{tr}\Big[e^{-\beta\hat{H}}\hat{F}e^{-\beta\hat{H}}e^{-i\hat{H}t/\hbar}h(s-\hat{x})e^{+i\hat{H}t/\hbar}\Big] \tag{2.7}$$

or equivalently

$$k(T)Q_r(T) = -\lim_{t\to\infty} \mathrm{tr}\Big[e^{-\beta\hat{H}}\hat{F}e^{-\beta\hat{H}}e^{-i\hat{H}t/\hbar}\hat{h}e^{+i\hat{H}t/\hbar}\Big] \tag{2.8}$$

define flux-side correlation function

$$c_{fs}(t) = \mathrm{tr}\Big[e^{-\beta\hat{H}}\hat{F}e^{-\beta\hat{H}}\hat{h}(t)\Big] = \mathrm{tr}\Big[e^{-\beta\hat{H}}\hat{F}e^{-\beta\hat{H}}e^{+i\hat{H}t/\hbar}\hat{h}e^{-i\hat{H}t/\hbar}\Big] \tag{2.9}$$

side-side correlation function

$$c_{ss}(t) = \mathrm{tr}\Big[e^{-\beta\hat{H}}\hat{h}e^{-\beta\hat{H}}(1-\hat{h}(t))\Big] \tag{2.10}$$

and flux-flux correlation function

$$c_{ff}(t) = \mathrm{tr}\Big[e^{-\beta\hat{H}}\hat{F}e^{-\beta\hat{H}}\hat{F}(t)\Big] \tag{2.11}$$

One can show that $c_{ss}(t)$ and $c_{ff}(t)$ are real and even functions of t, whereas $c_{fs}(t)$ is a real and odd function of t. Then the rate constant can be written as

$$k(T)Q_r(T) = \lim_{t\to\infty}\frac{d}{dt}c_{ss}(t) = \lim_{t\to\infty} c_{fs}(t) = \int_0^\infty dt\, c_{ff}(t) \tag{2.12}$$

define the Kubo-transformed time correlation function

$$\tilde{c}_{AB}(t) = \frac{1}{\beta}\int_0^\beta d\lambda\ \mathrm{tr}\Big[e^{-(\beta-\lambda)\hat{H}}\hat{A}e^{-\lambda\hat{H}}\hat{B}(t)\Big] \tag{2.13}$$

the rate coefficient can also be written as

$$\boxed{k(T) = \frac{1}{Q_r(T)}\lim_{t\to\infty}\tilde{c}_{fs}(t)} \tag{2.14}$$

which is exact, and generalises to three dimensional multi-particle systems. Note that $k(T)$ is independent of s and $\tilde{c}_{fs}(t\to 0_+) \sim t^{1/2} \to 0$.

Appendix A

Consider an operator $\hat{A}$ with the following eigenvalue equation

$$\hat{A}|n\rangle = a_n|n\rangle \tag{A.1}$$

Define operator $\hat{B}$ via

$$\hat{B} = \hat{U}\hat{A}\hat{U}^\dagger \tag{A.2}$$

where $\hat{U}$ is unitary. Then

$$\hat{B}(\hat{U}|n\rangle) = \hat{U}\hat{A}\hat{U}^\dagger\hat{U}|n\rangle = \hat{U}\hat{A}|n\rangle = a_n(\hat{U}|n\rangle) \tag{A.3}$$

thus $\hat{U}|n\rangle$ is an eigenket of $\hat{B}$ with eigenvalue a_n. A function of $\hat{A}$ by definition has the following spectral expansion

$$f(\hat{A}) = \sum_n |n\rangle f(a_n)\langle n| \tag{A.4}$$

then

$$\hat{U}f(\hat{A})\hat{U}^\dagger = \sum_n \hat{U}|n\rangle f(a_n)\langle n|\hat{U}^\dagger = f(\hat{U}\hat{A}\hat{U}^\dagger). \tag{A.5}$$